

The Pediatric Choroidal and Ciliary Body Melanoma Study

A Survey by the European Ophthalmic Oncology Group

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Purpose: To collect comprehensive data on choroidal and ciliary body melanoma (CCBM) in children and to validate hypotheses regarding pediatric CCBM: children younger than 18 years, males, and those without ciliary body involvement (CBI) have more favorable survival prognosis than young adults 18 to 24 years of age, females, and those with CBI.

Design: Retrospective, multicenter observational study.

Participants: Two hundred ninety-nine patients from 24 ocular oncology centers, of whom 114 were children (median age, 15.1 years; range, 2.7–17.9 years) and 185 were young adults.

Methods: Data were entered through a secure website and were reviewed centrally. Survival was analyzed using Kaplan-Meier analysis and Cox proportional hazards regression.

Main Outcome Measures: Proportion of females, tumor-node-metastasis (TNM) stage, cell type, and melanoma-related mortality.

Results: Cumulative frequency of having CCBM diagnosed increased steadily by 0.8% per year of age between 5 and 10 years of age and, after a 6-year transition period, by 8.8% per year from age 17 years onward. Of children and young adults, 57% and 63% were female, respectively, which exceeded the expected 51% among young adults. Cell type, known for 35% of tumors, and TNM stage (I in 22% and 21%, II in 49% and 52%, III in 30% and 28%, respectively) were comparable for children and young adults. Melanoma-related survival was 97% and 90% at 5 years and 92% and 80% at 10 years for children compared with young adults, respectively ($P = 0.013$). Males tended to have a more favorable survival than females among children (100% vs. 85% at 10 years; $P = 0.058$). Increasing TNM stage was associated with poorer survival (stages I, II, and III: 100% vs. 86% vs. 76%, respectively; $P = 0.0011$). By multivariate analysis, being a young adult (adjusted hazard rate [HR], 2.57), a higher TNM stage (HR, 2.88 and 8.38 for stages II and III, respectively), and female gender (HR, 2.38) independently predicted less favorable survival. Ciliary body involvement and cell type were not associated with survival.

Conclusions: This study confirms that children with CCBM have a more favorable survival than young adults 18 to 25 years of age, adjusting for TNM stage and gender. The association between gender and survival varies between age groups. *Ophthalmology* 2016;123:898-907 © 2016 by the American Academy of Ophthalmology.



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Uveal melanoma (UM) has an annual incidence of 2 to 8 per million in North America and Europe, varying by age, ethnicity, and latitude.¹⁻³ It is generally a disease of middle-aged and older adults, with a low incidence before 45 years

of age; the median age at diagnosis has increased to 62 years because of increasing life expectancy.^{3,4} Nevertheless, UM can occur at any age, even as a congenital tumor.^{5,6} Oculo(dermal) melanocytosis, neurofibromatosis type 1, familial